Şükrü Tarık Kostak

Design Engineer

+90 539 432 71 28

kostaks@itu.edu.tr

Work Experiences

Turkish Aerospace
May 2023 - Present
Ankara, Turkey

Structural Design Chief Engineer

- Team management (1 to 1, coaching and mentoring)
- Project management (work planning, distribution and schedule)
- Structural design (team lead, CDR and SDD documentation, design approvals, FPQ for composite and additive parts)
- System engineering (fire resistance test and structural design requirements, compliance matrices)
- R&D (UD carbon spar, additive perforated plate, dynamic waterjet, rubber seal design, electron beam welding)

Structural Design Engineer

March 2018 - May 2023

- Conceptual design (configuration selection trade offs, OML design)
- Structural design (structural layout and interface trade offs, SRR and PDR documentation)
- System installation (engine installation and engine structural mounting trade offs)
- Aft fuselage design (detail metallic, composite, sheet-metal parts and assembly release, retrofit for first flight)
- R&D (Python: PLM automation for CR forms, stacking optimization, image search engine, alternative fastener selector)

Hexagon Studio
September 2014 - August 2018
Kocaeli, Turkey

Intern

- Design (trim parts and wheel cover)
- Manufacturing (composite vacuum infusion parts, painting)

Projects

NATO SET-252
Design of a Validation Model of a Stealth UCAV, 10.14339/STO-TR-SET-252

Turkish Aerospace
Alternative Fastener Selection with Machine Learning
Aerospace Image Search Engine
Local Deepsearch Assistant for Public Company Folders

ITU Facilis Student Team
Racing Car for Shell Eco-marathon
Responsible for steering, transmission, carbon fiber monocoque body design, test systems, and ECU calibration.
Modelled race strategy via Simulink, developed 3D-printed parts, and optimized fuel consumption through engine mapping.

Graduation Project
Dynamic Modeling of Lightweight Gasoline Vehicles and Track Simulation

Patents

Embedded Ejector System for Fighter Jets Engine Bay (In review)

Porous Ventilation Inlet and Exhaust for Fighter Jets Engine Bay (In preparation)

Publications

Ven, H. van der, Bocanegra, D. E., González, J. Álvarez ., İnal, M. E. ., Altinoklu, A., Öztürk, A. K., Jakobus, U., Osipov , A., Lie-Svendsen, Øystein, Weinmann, F., Andersson, Åsa, Edefur, H. ., Hall, J.-O., Martínez, D. P., Çiftçi, T., & Kostak, Şükrü T. (2024). **Design of a Radar Signature Measurement Model of an Unmanned Aerial Vehicle with Low Radar Signature.** Applied Computational Electromagnetics Society Journal (ACES), 39(06), 478–489. https://doi.org/10.13052/2024.ACES.J.390602

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Education

Istanbul Technical University
2013 - 2018
Istanbul, Turkey

Bsc. Mechanical Engineering
GPA: 2.98

Certificates

CADEM	CATIA Composite Part Design (CPD) Module
CADEM	CATIA ICEM Shape Design Module
Turkish Aerospace	Leadership Certification

Language

Turkish	Native
English	Advanced

Software & Tools

MATLAB
Simulink
CATIA v5
Python
VBA

Siemens NX
SOLIDWORKS
CURA
3D Experience
Part Design | Composite Part Design | Hydroformed Sheet Design | Lattice Design | Process Composer | Function Driven Generative Design | Visual Scripting | Mechanical Scenario | Physics Results

Key Areas of Expertise

Structural & Conceptual Design

Aircraft Conceptual Design | Composite Materials & Laminate Optimization | Structural Testing | Structural Optimization | Rubber Seal Design and Testing | Gyroid Lattice Structure Design | Tolerance Chain Analysis | Low Observable Design

Computational Engineering & Automation

Computational Design | Parametric Design & Process Automation (VBA, Python) | Digital Twin & Model-Based Systems Engineering | Technical Documentation

AI & Data-Driven Engineering

AI & Machine Learning in Engineering Workflows | Local LLM-based Document Analysis & RAG | Multidisciplinary Design Optimization

Innovation & Leadership

Intrapreneurship | Innovation | Team Performance | Behavioral Diagnostics

Interests

Computational Design & AI in Engineering
Agile Problem Solving in Aerospace
Cognitive Science & Scientific Skepticism
Philosophy of Consciousness & Decision-Making
Technical Writing & Info-graphics
Innovation & Patent Research
Workflow & Performance Optimization
Chronotype-Based Team Productivity
Leadership & Team Development

Soft Skills

Communication & Collaboration

- Effective Communication
- Teamwork
- Active Listening
- Persuasion
- Cross-Cultural Awareness
- Empathy

Critical Thinking & Decision Making

- Problem Solving
- Critical Thinking
- Creative Thinking
- Decision Making

Self-Management & Discipline

- Time management
- Self-Motivation
- Self-Discipline
- Initiative
- Willingness to Learn

Leadership & Responsibility

- Leadership
- Conflict Resolution
- Composure Under Pressure